Université d'Ottawa · University of Ottawa

Faculté des sciences
Mathématiques et de statistique

Faculty of Science
Mathematics and Statistics

MAT 1302 B: Mathematical methods II

Instructor: Aziz Khanchi

Final Exam
April 2011

Surname _____ First Name _____

Student # _____ Seat # _____

Instructions:

- No calculators or other electronic aids allowed.
- No notes, books or other papers allowed.
- Answer all questions in the space provided. You must justify your answers and explain your reasoning.
- In all there are 17 questions worth a total of 55 marks. There are 8 multiple choice questions, and 9 long answer questions. There will be no part marking for the multiple choice questions. For the long answer questions, you must justify your answers.

GOOD LUCK!

Answers to multiple choice Qs

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8

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Student # _____ MAT 1302 B Final Exam

Do not write anything in this table.

Q1-Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Total
/16	/3	/3	/5	/4	/7	/3	/3	/ 6	/5	/55

Part 1: Multiple choice questions, 2 points each (no partial marks).

Question 1. For which values of h and k does the system

$$\begin{aligned} x_1 + 3x_2 &= k \\ 4x_1 + hx_2 &= 8 \end{aligned}$$

have no solutions?

A) $h = 12,\ k = 2$

✓**B)** $h = 12,\ k \neq 2$

C) $k \neq 2$ and $h \in \mathbb{R}$

D) For all $h, k \in \mathbb{R}$

E) For all values of k and h the system has solutions.

$$\left[\begin{array}{cc|c} 1 & 3 & k \\ 4 & h & 8 \end{array}\right] \xrightarrow{-4R_1+R_2 \to R_2} \left[\begin{array}{cc|c} 1 & 3 & k \\ 0 & h-12 & 8-4k \end{array}\right]$$

inconsistent if $h-12=0$ and $8-4k\neq 0$

Question 2. Let $A = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 0 & 2 \\ 4 & 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, A^TB is

A) Not defined

B) $\begin{bmatrix} 11 \\ 7 \\ 4 \end{bmatrix}$

C) $\begin{bmatrix} 4 \\ 7 \\ 11 \end{bmatrix}$

✓**D)** $\begin{bmatrix} 15 \\ 2 \\ 6 \end{bmatrix}$

E) $\begin{bmatrix} 1 & 1 & 4 \\ 2 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$

$$A^TB = \begin{bmatrix} 1 & 1 & 4 \\ 2 & 0 & 0 \\ 2 & 2 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 15 \\ 2 \\ 6 \end{bmatrix}$$

Question 3. Let $\vec{u}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$, $\vec{u}_2 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$, $\vec{u}_3 = \begin{bmatrix} -2a \\ b \\ -a \end{bmatrix}$ be vectors in $\mathbb{R}^3$. Under which condition $\{\vec{u}_1, \vec{u}_2, \vec{u}_3\}$ is a basis for $\mathbb{R}^3$?

A) $a = 0,\ b \in \mathbb{R}$

✓**B)** $a \neq b$.

C) $a \neq -b$.

D) $a = -b$.

E) $a = b$.

These form a basis is $[\vec{u}_1\ \vec{u}_2\ \vec{u}_3]$ is invertible

or $\begin{vmatrix} 1 & 2 & -2a \\ -1 & -1 & b \\ 0 & 1 & -a \end{vmatrix} \neq 0$

$$\begin{vmatrix} 1 & 2 & -2a \\ -1 & -1 & b \\ 0 & 1 & -a \end{vmatrix} = \begin{vmatrix} 1 & 2 & -2a \\ 0 & 1 & -2a+b \\ 0 & 1 & -a \end{vmatrix} = (-1)^2(1)\begin{vmatrix} 1 & -2a+b \\ 1 & -a \end{vmatrix} = -a+2a-b = a-b \neq 0$$

$\Rightarrow a \neq b$

Question 4. Suppose A, B and C are 3×3 matrices with $\det A = -1$, $\det B = 2$, $\det C = 3$. Find $\det(2A^5C^TB^TC^{-1})$.

✓**A)** -16

B) 16

C) -4

D) 4

E) -36

$$2^3(-1)^5 \cdot 3 \cdot 2 \cdot \frac{1}{3} = -2^4 = -16$$

Question 5. Let $B = \begin{bmatrix} 4 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & 2 & 4 \end{bmatrix}$, then

A) B has 3 distinct eigenvalues and $detB$=16

✓**B)** B has 2 distinct eigenvalues and $detB$=16

C) B has 3 distinct eigenvalues and $detB$=9

D) B has 1 distinct eigenvalue and $detB$=16

E) B has 2 distinct eigenvalues and $detB$=9

evalues: 4, 1

$detB = 4 \cdot 4 \cdot 1 = 16$

Question 6. Which of the following statements are true?

1. It is possible for a set of 4 vectors in $\mathbb{R}^3$ to be linearly independent.

2. Any basis of $\mathbb{R}^5$ must consist of 5 vectors. ✓

3. It is possible for a set of 4 vectors to span $\mathbb{R}^3$ ✓

4. Any basis of a subspace contains the zero vector.

5. A subspace can have more than one basis. ✓

✓**A)** Statements 2, 3, 5

B) Statements 1, 2, 4, 5

C) Statements 2, 3, 4, 5

D) Statements 1, 4, 5

E) All of the statements are False!

Question 7. Which sets are linearly independent?

Three vectors in $\mathbb{R}^2$

$$A = \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 6 \\ 1 \end{bmatrix} \right\}, \quad B = \left\{ \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\},$$

dependent because of zero vector

$$C = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 8 \end{bmatrix} \right\}, \quad D = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

non-zero vector

A) A and B.

√**B)** C and D.

C) A and C.

D) All of them.

E) None of them.

$$\begin{bmatrix} 1 & 4 \\ 0 & 0 \\ 1 & 8 \end{bmatrix} \longrightarrow \begin{bmatrix} \textcircled{1} & 4 \\ 0 & 0 \\ 0 & \textcircled{4} \end{bmatrix}$$

pivots in each column

Question 8. Which of the following statements are true for an $n \times n$ matrix A ?

1. $\det A = \det A^T$. ✓

2. For every $\vec{b} \in \mathbb{R}^n$, the equation $A\vec{x} = \vec{b}$ has a solution. ✗

3. The equation $A\vec{x} = 0$ always has a solution. ✓

4. $\dim(\text{Nul}A) \leq n$. ✓

5. Rank$A$=n. ✗

√**A)** Statements 1, 3, 4

B) Statements 3, 4, 5

C) Statements 1, 2, 4

D) Statements 1, 4, 5

E) All of the statements are True

Long Answer Questions:

Question 9. [3 marks] Consider the production model $\vec{x} = C\vec{x} + \vec{d}$ for an economy with two sectors, where the consumption matrix is

$$C = \begin{bmatrix} 0 & 0.5 \\ 0.6 & 0.2 \end{bmatrix}$$

Determine the production level necessary to satisfy the final demand of $\vec{d} = \begin{bmatrix} 10 \\ 20 \end{bmatrix}$.

Solve $(I-C)\vec{x} = \vec{d} \longrightarrow \vec{x} = (I-C)^{-1}\vec{d}$

$$(I-C) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0.5 \\ 0.6 & 0.2 \end{bmatrix} = \begin{bmatrix} 1 & -0.5 \\ -0.6 & 0.8 \end{bmatrix}$$

$$(I-C)^{-1} = \frac{1}{0.8-0.3}\begin{bmatrix} 0.8 & 0.5 \\ 0.6 & 1 \end{bmatrix} = \begin{bmatrix} 1.6 & 1 \\ 1.2 & 2 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 1.6 & 1 \\ 1.2 & 2 \end{bmatrix}\begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{bmatrix} 16+20 \\ 12+40 \end{bmatrix} = \begin{bmatrix} 36 \\ 52 \end{bmatrix}$$

Question 10. [3 marks] Let $z = 2 - 2i$, $w = -1 + i$ and $r = -2$. Calculate the following.

A) $\frac{\overline{z}}{w} = \frac{2+2i}{-1+i} = \frac{2+2i}{-1+i}\,\frac{-1-i}{-1-i} = \frac{(-2+2)+(-2-2)i}{(-1)^2+1^2} = \frac{-4i}{2} = -2i$

B) $|rz| = |-2||z| = 2\sqrt{2^2+(-2)^2} = 2\sqrt{4+4} = 2\sqrt{8}$

Question 11. [5 marks] Let $A = \begin{bmatrix} 1 & 3 & 1 \\ -4 & -9 & 2 \\ 0 & -3 & -6 \end{bmatrix}$.

A) Solve the homogeneous system $A\vec{x} = 0$.

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ -4 & -9 & 2 & 0 \\ 0 & -3 & -6 & 0 \end{array}\right] \xrightarrow{4R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & 3 & 6 & 0 \\ 0 & -3 & -6 & 0 \end{array}\right] \xrightarrow[R_2+R_3 \to R_3]{-R_2+R_1 \to R_1}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -5 & 0 \\ 0 & 3 & 6 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{\frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & 0 & -5 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

x_3 free

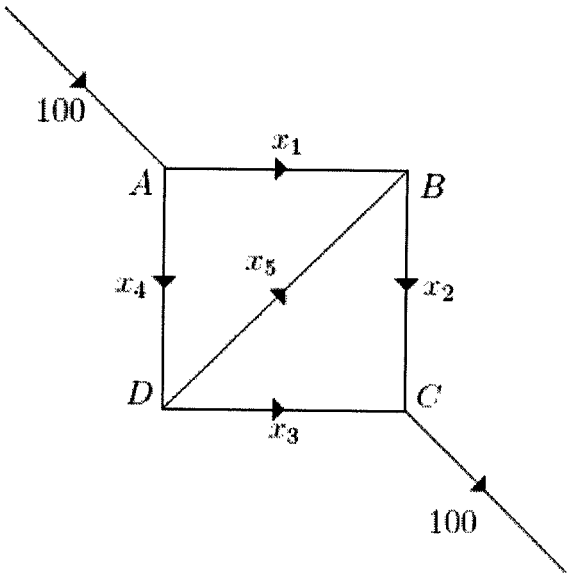
$$\begin{cases} x_1 = 5t \\ x_2 = -2t \\ x_3 = t \end{cases} \qquad \vec{x} = \begin{bmatrix} 5t \\ -2t \\ t \end{bmatrix}$$

B) Provide vectors that span the solution set of $A\vec{x} = 0$.

$$\vec{x} = t\begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$$

$$\text{Solution set} = \text{span}\left\{\begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}\right\}$$

Question 12. [4 marks] The traffic flow in a city is represented by the diagram below. The arrows indicate the direction of one-way traffic.



A) Write down a linear system describing this traffic flow. Do not perform any calculations at this stage.

Node	Flow in	Flow out
A	$100 = x_1 + x_4$	
B	$x_1 + x_5 = x_2$	
C	$x_2 + x_3 = 100$	
D	$x_4 = x_3 + x_5$	
Overall	$100 = 100$	

B) The reduced echelon form of the augmented matrix in part (a) is

$$\begin{array}{c} \\ \\ \\ \\ \\ \end{array}\!\!\begin{array}{c} \\ \\ \\ x_4 \\ \\ \end{array}\!\!\begin{array}{c} \\ \\ \\ x_5 \\ \\ \end{array}\!\!\begin{array}{c} \\ \\ \\ \text{free} \\ \\ \end{array}$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 1 & 0 & 100 \\ 0 & 1 & 0 & 1 & -1 & 100 \\ 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Give the general solution of the system (ignore the constraints on the variables for now).

$$\begin{cases} x_1 = 100 - x_4 \\ x_2 = 100 - x_4 + x_5 \\ x_3 = x_4 - x_5 \\ x_4, x_5 \text{ free} \end{cases}$$

C) If road DB is closed due to construction and if the flow on road AD is $x_4 = 10$, what is the traffic flow on road BC?

The two conditions give $x_4 = 10$ and $x_5 = 0$. Substituting these values into the general solution, we obtain

$$x_2 = 100 - 10 = 90.$$

Question 13. [7 marks] Let $A = \begin{bmatrix} 4 & 0 & -2 \\ 2 & 5 & 4 \\ 0 & 0 & 5 \end{bmatrix}$.

A) Find the characteristic polynomial of A.

$$|A-\lambda I| = \begin{vmatrix} 4-\lambda & 0 & -2 \\ 2 & 5-\lambda & 4 \\ 0 & 0 & 5-\lambda \end{vmatrix} = (4-\lambda)\begin{vmatrix} 5-\lambda & 4 \\ 0 & 5-\lambda \end{vmatrix} - 2\begin{vmatrix} 2 & 5-\lambda \\ 0 & 0 \end{vmatrix}$$

$$= (4-\lambda)(5-\lambda)^2$$

B) List the eigenvalues and their multiplicities.

$\lambda = 4$ with multiplicity 1

$\lambda = 5$ " " 2

C) Describe all the corresponding eigenvectors.

For $\lambda = 4$: $[A-4I|\vec{0}] = \left[\begin{array}{ccc|c} 0 & 0 & -2 & 0 \\ 2 & 1 & 4 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \longrightarrow$

$$\left[\begin{array}{ccc|c} 1 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \begin{cases} x_1 = \frac{-1}{2}x_2 \\ x_2 \text{ free} \\ x_3 = 0 \end{cases} \quad \begin{bmatrix} \frac{-1}{2}x_2 \\ x_2 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} -1/2 \\ 1 \\ 0 \end{bmatrix}$$

For $\lambda = 5$: $[A-5I|\vec{0}] = \left[\begin{array}{ccc|c} -1 & 0 & -2 & 0 \\ 2 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$$\begin{cases} x_1 = -2x_3 \\ x_2 \text{ free} \\ x_3 \end{cases} \qquad \begin{bmatrix} -2x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

D) If possible, diagonalize the matrix A. That is, find a diagonal matrix D and invertible matrix P such that $A = PDP^{-1}$. (Note: You are not asked to find P^{-1}.)

$$D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} \qquad P = \begin{bmatrix} -1 & 0 & -2 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 14. [3 marks] Is the matrix $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 4 \end{bmatrix}$ invertible? If so, find A^{-1}.

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 2 & 1 & 4 & 0 & 0 & 1 \end{array}\right] \xrightarrow[-2R_1+R_3 \to R_3]{-R_1+R_2 \to R_2}$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & -1 & 1 & 0 \\ 0 & -1 & 0 & -2 & 0 & 1 \end{array}\right] \xrightarrow[-R_3+R_2 \to R_2]{R_3+R_1 \to R_1}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & 0 & +1 \\ 0 & 0 & -1 & +1 & 1 & -1 \\ 0 & -1 & 0 & -2 & 0 & 1 \end{array}\right] \xrightarrow{2R_2+R_1 \to R_1}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & -1 \\ 0 & 0 & -1 & +1 & 1 & -1 \\ 0 & -1 & 0 & -2 & 0 & 1 \end{array}\right] \xrightarrow[-R_3]{-R_2}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -1 & -1 & 1 \\ 0 & 1 & 0 & 2 & 0 & -1 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_3} \underbrace{\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right.}_{I} \left.\begin{array}{ccc} 1 & 2 & -1 \\ 2 & 0 & -1 \\ -1 & -1 & 1 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & -1 \\ -1 & -1 & 1 \end{bmatrix}$$

Question 15. [3 points] Find the determinant of $A = \begin{bmatrix} 2 & -4 & 6 & 1 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 0 \\ 1 & -1 & 1 & 0 \end{bmatrix}$.

$$\begin{vmatrix} 2 & -4 & 6 & 1 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 0 \\ 1 & -1 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 2 & -4 & 6 & 1 \\ -3 & 10 & -12 & 0 \\ 2 & 3 & 2 & 0 \\ 1 & -1 & 1 & 0 \end{vmatrix} =$$

$-2R_1+R_2$

$$(-1)^{1+4}(1)\begin{vmatrix} -3 & 10 & -12 \\ 2 & 3 & 2 \\ 1 & -1 & 1 \end{vmatrix} = -\begin{vmatrix} -3 & 10 & -12 \\ 0 & 5 & 0 \\ 1 & -1 & 1 \end{vmatrix}$$

$-2R_3+R_2$

$$= -(-1)^{2+2}(5)\begin{vmatrix} -3 & -12 \\ 1 & 1 \end{vmatrix} = (-1)(1)5(-3+12)$$

$$= (-5)(9) = -45$$

Question 16. [6 marks] Consider the matrix $A = \begin{bmatrix} 1 & 2 & -5 & 0 & -1 \\ 2 & 5 & -8 & 4 & 3 \\ -3 & -9 & 9 & -7 & -2 \\ 3 & 10 & -7 & 11 & 7 \end{bmatrix}$, which can be reduced to $\begin{bmatrix} \textcircled{1} & 2 & -5 & 0 & -1 \\ 0 & \textcircled{1} & 2 & 4 & 5 \\ 0 & 0 & 0 & \textcircled{1} & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

A) Find a basis for $ColA$. Columns 1, 2 & 4 are pivot.

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ -3 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ -9 \\ 10 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \\ -7 \\ 11 \end{bmatrix} \right\}$$

B) What is the $RankA$? 3

C) Find a basis for $NulA$. Solve $A\vec{x} = \vec{0}$

$$A \sim \begin{bmatrix} 1 & 2 & -5 & 0 & -1 \\ 0 & 1 & 2 & 4 & 5 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[-4R_3+R_2 \to R_2]{-2R_2+R_1 \to R_1} \begin{bmatrix} 1 & 0 & -9 & -8 & -11 \\ 0 & 1 & 2 & 0 & -3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{8R_3+R_1 \to R_1} \begin{bmatrix} 1 & 0 & -9 & 0 & 5 \\ 0 & 1 & 2 & 0 & -3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

General solution

$$\begin{cases} x_1 = 9x_3 - 5x_5 \\ x_2 = -2x_3 + 3x_5 \\ x_4 = -2x_5 \\ x_3 \text{ \& } x_5 \text{ free} \end{cases}$$

$$\begin{bmatrix} 9x_3 - 5x_5 \\ -2x_3 + 3x_5 \\ x_3 \\ -2x_5 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} 9 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -5 \\ 3 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

D) What is the dimension $NulA$? 2

E) State the equation relating the number of columns of a matrix, rank and the dimension of the null space.

rank A + dim Nul A = # of columns

3 + 2 = 5

Basis for nul A =

$$\left\{ \begin{bmatrix} 9 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ 3 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

Question 17. [5 marks] There are 20 cockroaches living in a suite of a five star hotel. The cockroaches move between the kitchenette and bathroom. At the beginning, there are 10 cockroaches in the kitchenette and 10 in the bathroom. Every week 40% of the cockroaches in the kitchenette go to the bathroom and 30% of the cockroaches in the bathroom go to the kitchen.

A) Determine the migration matrix associated with this model.

$$M = \begin{matrix} \\ K \\ B \end{matrix}\begin{matrix} \begin{matrix} K & \quad B \end{matrix} \\ \begin{bmatrix} 0.6 & 0.3 \\ 0.4 & 0.7 \end{bmatrix} \end{matrix}$$

B) Compute the expected number of cockroaches in the bathroom after 2 weeks.

$$\vec{x}_0 = \begin{bmatrix} 10 \\ 10 \end{bmatrix}, \quad \vec{x}_1 = M\vec{x}_0 = \begin{bmatrix} 0.6 & 0.3 \\ 0.4 & 0.7 \end{bmatrix}\begin{bmatrix} 10 \\ 10 \end{bmatrix} = \begin{bmatrix} 6+3 \\ 4+7 \end{bmatrix} = \begin{bmatrix} 9 \\ 11 \end{bmatrix}$$

after one week

$$\vec{x}_2 = M\vec{x}_1 = \begin{bmatrix} 0.6 & 0.3 \\ 0.4 & 0.7 \end{bmatrix}\begin{bmatrix} 9 \\ 11 \end{bmatrix} = \begin{bmatrix} 5.4+3.3 \\ 3.6+7.7 \end{bmatrix} = \begin{bmatrix} 8.7 \\ 11.3 \end{bmatrix}$$

Expected # in the Kitchen 8.7, expected # in the Bathroom 11.3

C) After the migration goes on for a long period of time, find the proportion of the cockroaches in the kitchenette and bathroom.
(Hint: Find the steady-state vector of the migration matrix.)

Solve $(M-I)\vec{x} = \vec{0}$ $\quad \left[\begin{array}{cc|c} -0.4 & 0.3 & 0 \\ 0.4 & -0.3 & 0 \end{array}\right] \longrightarrow$

Sol:

$$\left[\begin{array}{cc|c} 1 & \frac{-3}{4} & 0 \\ 0 & 0 & 0 \end{array}\right] \qquad \begin{bmatrix} \frac{3}{4}x_2 \\ x_2 \end{bmatrix}$$

$$\frac{3}{4}x_2 + x_2 = 1 \longrightarrow x_2 = \frac{4}{7}$$

Steady-state $\begin{bmatrix} 3/7 \\ 4/7 \end{bmatrix}$ $\quad \frac{3}{7}$ of the population will be in K.

$\frac{4}{7}$ " " " " " " B.

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